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$$f(x) = x^{\sin x}$$

$$\Rightarrow y = x^{\sin x}$$

$$\Rightarrow \ln y = \ln x^{\sin x} = (\sin x) \ln x$$

$$\Rightarrow (\ln y)' = ((\sin x) \ln x)'$$

$$\Rightarrow \frac{y'}{y} = ((\cos x) \ln x) + \frac{\sin x}{x}$$

$$\Rightarrow y' = y \left(((\cos x) \ln x) + \frac{\sin x}{x} \right)$$

$$\Rightarrow y' = x^{\sin x} (\cos x) \ln x + x^{\sin x - 1} \sin x$$

$$\Rightarrow \boxed{f'(x) = x^{\sin x} (\cos x) \ln x + x^{\sin x - 1} \sin x}$$

References